APERIO | AUTONOMOUS EARTH INTELLIGENCE PLATFORM

AUTHORITY DISASTER INTELLIGENCE ASSESSMENT • DEFORESTATION

Location: Forest Grid (10.89N, 76.52E), Kerala, India • Generated: 08 Oct 2026 04:14 UTC • Status: OPERATOR VERIFIED

Target Event	Deforestation • Forest Grid (10.89N, 76.52E)	**Risk Probability**	**32.0% (HIGH)**
Affected Area	7.03 km²	**Model Confidence**	92.0% Indicative
Exposed Population	4,250	**Antecedent Rainfall**	180.0 mm (72h: 572mm)
Terrain Slope	35.0° (Steep escarpment)	**SAM Scar Boundary**	Validated (7.03 km²)
Satellite Observation	Sentinel-1 SAR + Sentinel-2 MSI	**Human Verification**	OPERATOR VERIFIED

Multi-Agent Intelligence & Evidence Progression

[SYNTHETIC DEMO DATA NOTICE] This report record is derived from synthetic demonstration data (*aperio_synthetic_deforestation_dataset.csv*) and does not represent verified satellite ground truth.
1. Satellite Optical Ingestion: Simulated Sentinel-2 multi-spectral canopy reflectance with NDVI change calculation.
2. SAM Segmentation: Simulated Zero-Shot Segment Anything canopy loss mask.
3. XGBoost & Bayesian Sequential Update: Optical disturbance and SAR backscatter features updated beliefs to calibrated probability rating of **32.0%**.

Conservation Advisory & Ground Inspection Planning

Designation	Unit / Command	Distance / Status	Exposure
State Forest Division Monitoring Post (Kerala)	Forestry Range Unit	1.5 km • Patrol Track Open	**MONITORING**
Forest Grid (10.89N, 76.52E) Buffer Perimeter Station	Conservation Squad	3.2 km • Operational	**SURVEILLANCE**

Accountable Operator Review & Audit Trail

Reviewing Desk: APERIO Response Operations Desk
Operator Notes: Operator confirmed sensor convergence. Priority advisory drafted. Emergency services notified of bridge disruption.
Operational Boundary: Decision Support Output only. All evacuations require local ground confirmation.